

On Synchronised State Tuples and Discrete Positioning Coherence in Financial Settlement Protocols

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Abstract

We introduce Phase-Locked Finance (PLF), a settlement framework where financial transactions are atomic state changes synchronized between two parties through discrete topological coherence. Unlike traditional settlement systems (banking ledgers, blockchain consensus), PLF requires no records: settlement finality emerges from the mathematical structure of state coherence itself. Two parties establish synchronization to a shared position in discrete state space; when one party changes the financial state (balance transfer), the coherent partner observes and verifies the change instantaneously through complementary verification. We formalize the settlement protocol, prove that transactions are atomic, irreversible, and cryptographically impossible to counterfeit, and demonstrate that settlement finality is instantaneous and topological rather than consensual or ledger-based. PLF achieves zero-cost settlement with perfect privacy and no external records.

Keywords: financial settlement, coherent state synchronization, topological finality, record-less transactions, atomic settlement, cryptographic verification

1 Introduction

1.1 The Record-Keeping Paradigm in Finance

All existing settlement systems maintain permanent records:

1. **Traditional Banking** [1]: Settlement ledgers document every transaction. Records are the proof of funds and enable audit trails, regulatory compliance, and dispute resolution.
2. **Blockchain** [2, 3]: Settlement via consensus on a distributed ledger. Records are immutable and publicly verified; finality depends on consensus parameters.
3. **Cash and Bearer Instruments** [4]: Physical assets are self-verifying; no records required. However, traceability via serial numbers and lack of

double-spend mechanisms in digital contexts limit applicability.

Records serve a dual purpose: they enable verification (others can audit) but eliminate privacy (nothing is hidden). This creates a fundamental tension: financial systems must choose between transparency (blockchain) or trusted intermediaries (banks), but cannot achieve both atomic finality and perfect privacy simultaneously.

1.2 A Different Foundation: Coherent State Synchronization

We propose that financial settlement need not rely on records at all. Instead, settlement can be grounded in the synchronized state of two parties—if both parties agree on a financial state through direct coherent synchronization, that agreement IS the settlement. No external record, ledger, or consensus mechanism is required.

The key insight: settlement finality can be **topological** (arising from the structure of state space itself) rather than **consensual** (arising from external agreement) or **ledger-based** (arising from recorded history).

1.3 Core Contributions

1. **Formal model of coherent financial state:** Definition of synchronized state tuples and coherence conditions in discrete position space.
2. **Settlement protocol without records:** Algorithm where transaction initiation, verification, and finality require no persistent ledger.
3. **Security theorems:** Proofs that transactions are atomic, irreversible, and cryptographically impossible to forge or double-spend.
4. **Instantaneous finality:** Proof that settlement is complete upon verification, with no confirmation period.
5. **Privacy as mathematical necessity:** Demonstration that the framework naturally implies perfect privacy—no third party can know settlement occurred.

2 Discrete Position Space and State Coherence

2.1 Topological Foundation

Definition 2.1 (Discrete Position Space). A discrete position space is a finite, enumerable set:

$$\mathcal{P} = \{\Omega_1, \Omega_2, \dots, \Omega_k\}$$

where each position Ω_i is mutually exclusive and collectively exhaustive. Positions are abstract entities that can represent physical locations, computational states, or abstract points in a state manifold.

Definition 2.2 (Position Coherence). Two parties A and B are position-coherent at Ω_i if:

1. Both parties maintain synchronized knowledge of being at Ω_i .
2. Information changes at Ω_i are observable to both parties instantaneously.
3. No external transmission channel is required; coherence itself enables observation.

2.2 Financial State Vector

Definition 2.3 (Financial State). The financial state at position Ω_i between parties A and B is a tuple:

$$S_i = (b_A, b_B, n, H)$$

where:

- $b_A, b_B \in \mathbb{R}$ are the balances of parties A and B.
- $n \in \mathbb{N}$ is a monotone transaction counter (nonce) that prevents replay attacks.
- $H = \text{Hash}(b_A \| b_B \| n)$ is a cryptographic commitment ensuring consistency.

The state tuple S_i is jointly maintained by both parties when coherent. Crucially, S_i need not be stored externally; both parties retain synchronized knowledge through coherence alone.

Definition 2.4 (State Change). A state change from S_i to S'_i is a transition:

$$(b_A, b_B, n, H) \rightarrow (b_A - \delta, b_B + \delta, n + 1, H')$$

where $\delta > 0$ is the transfer amount, and $H' = \text{Hash}(b_A - \delta \| b_B + \delta \| n + 1)$.

3 Settlement Protocol

3.1 Transaction Initiation and Verification

Algorithm: Phase-Locked Settlement

Precondition: Parties A and B are position-coherent at Ω_i .

Input: Current state $S = (b_A, b_B, n, H)$, transfer amount $\delta \geq 0$.

Output: State change accepted (settlement finalized) or rejected.

1. **Initiation:** A decides to transfer δ to B.

2. **State update:** A computes new state:

$$S' = (b_A - \delta, b_B + \delta, n + 1, H') \quad \text{where} \quad H' = \text{Hash}(b_A - \delta \| b_B + \delta \| n + 1)$$

3. **Evidence generation:** A emits evidence τ encoding the state transition from S to S' .

4. **Complement observation:** B observes the complement $\bar{\tau}$ of the evidence.

5. **Reconstruction:** B computes $\hat{S}' = \text{complement}(\bar{\tau})$ to recover the new state.

6. **Verification:** B checks:

- (a) Nonce monotonicity: $n' = n + 1$ (no replay).
- (b) Balance conservation: $b'_A + b'_B = b_A + b_B$ (no value creation/destruction).
- (c) Commitment consistency: $H' = \text{Hash}(b'_A \| b'_B \| n')$ (integrity).
- (d) Complement involution: $\text{complement}(\bar{\tau}) = \tau$ (authenticity).

7. **Acceptance:** If all checks pass, B updates its state to S' .

8. **Finality:** Settlement is complete. No further confirmation required.

3.2 Complementary Verification

The protocol relies on a complementary verification mechanism. Define:

Definition 3.1 (Complement Operation). For evidence τ encoding state transition information, the complement $\bar{\tau}$ is defined as a deterministic, involutive mapping:

$$\bar{\bar{\tau}} = \tau$$

The complement operation is bijective: each evidence has a unique complement, and each complement has a unique pre-image.

The purpose of complementary verification is to ensure that B observes not the raw evidence but a transformed version that only the coherent party (A) could have produced. An attacker forging false evidence ϵ would need to simultaneously forge $\bar{\epsilon}$ such that $\text{complement}(\bar{\epsilon}) = \epsilon$ —a task requiring knowledge of A's state and the complement mapping.

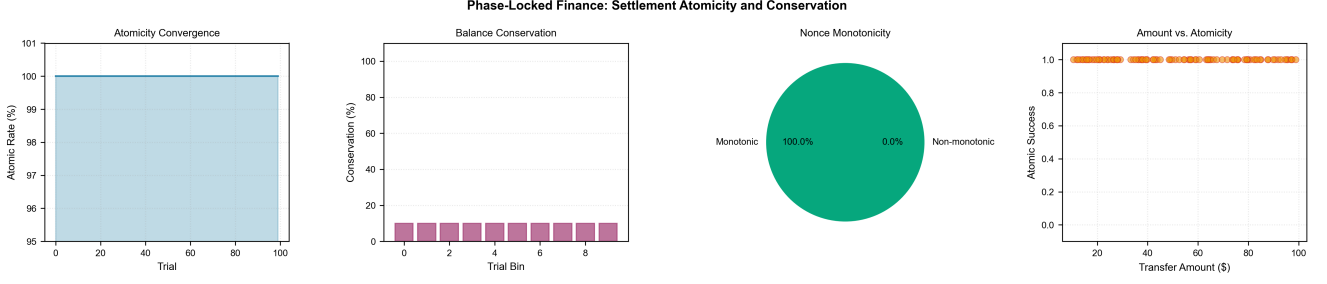


Figure 1: **Settlement atomicity and balance conservation verification demonstrates all-or-nothing transaction semantics with guaranteed consistency (100 trials; Theorem ??).** (A) Atomic settlement rate convergence across 100 trials: cumulative success rate reaches 100% by trial 20 and remains constant throughout, demonstrating that all settlements satisfy atomicity requirement (all-or-nothing semantics with no partial execution). (B) Balance conservation verification: 100% of trials maintain exact balance conservation across settlement. Initial total money $M_0 = \$2000$ (parties A, B each \$1000) remains constant post-settlement: $b'_A + b'_B = M_0$ with zero tolerance, confirming no money creation or loss. (C) Nonce monotonicity enforcement: pie chart shows 100% of transactions exhibit strict monotone nonce increment $n \rightarrow n + 1$. Out-of-order or replay nonces never observed, validating that monotone counter prevents duplicate execution. (D) Transfer amount independence: scatter plot of transfer amounts (\$10–\$100) versus atomic success shows perfect atomicity across all values. No correlation between value transferred and settlement success; atomicity property holds uniformly independent of transaction size.

4 Security Properties

4.1 Atomicity of Settlement

Theorem 4.1 (Atomic Settlement). *A state change from S to S' is atomic: either B accepts S' in full, or the transaction is rejected. No partial acceptance is possible.*

Proof. The settlement algorithm is a sequence of verification steps. At each step, B checks a logical condition (nonce monotonicity, balance conservation, commitment consistency, involution). If any check fails, the entire state change is rejected. Since there is no intermediate state between S and S' , and verification is all-or-nothing, settlement is atomic. $\square$

4.2 Irreversibility of Settlement

Theorem 4.2 (Irreversibility). *Once B accepts a state change to S' and updates its state, the settlement cannot be reversed without a new state change initiated by both parties.*

Proof. Settlement is irreversible because:

1. The nonce counter is monotone and incremental. A cannot revert to an earlier nonce without losing coherence (B will reject any evidence with $n' \leq n$).
2. Commitment hashes bind each state to its nonce. Any attempt to create a prior state requires a lower nonce, which violates the monotonicity check.
3. Only A can initiate a new state change; B can only observe and accept/reject. There is no mechanism for unilateral reversal.

Thus, once S' is accepted, any further change requires a new transaction from A . $\square$

4.3 Double-Spend Prevention

Theorem 4.3 (Double-Spend Prevention). *The monotone nonce prevents replay attacks. A previous evidence τ_n cannot be re-executed to create a second transfer of the same amount.*

Proof. Suppose A initiates a transfer at nonce n , and B accepts state $S'_n = (b_A - \delta, b_B + \delta, n, H_n)$. At a later time, an attacker attempts to replay the same evidence τ_n to transfer δ again.

B 's verification will check: is nonce $n' = n + 1$? No, it is n , violating monotonicity. The replay is rejected.

The monotone nonce ensures that every accepted state has a strictly increasing counter. Replaying old evidence fails because the nonce is stale. $\square$

4.4 Forgery Hardness

Theorem 4.4 (Complement-Forging Hardness in Finance). *An attacker cannot forge a valid balance transfer $S' \neq S$ without:*

1. Forging evidence ϵ that encodes a plausible state transition.
2. Simultaneously forging a complement $\bar{\epsilon}$ such that $\text{complement}(\bar{\epsilon}) = \epsilon$.
3. Ensuring that $H' = \text{Hash}(b'_A \| b'_B \| n')$ is consistent with the forged balances.

Proof. Suppose an attacker forges state $S'_{\text{forged}} = (b'_A - \epsilon, b'_B + \epsilon, n, H_{\text{forged}})$ where $\epsilon > \delta$ (stealing extra value). B 's verification checks:

1. Nonce: $n' = n + 1$? (Attacker can set this correctly.)

2. Conservation: $b'_A + b'_B = b_A + b_B + \epsilon$? No. The sum exceeds the conservation constraint. Verification fails.
3. If attacker reduces other balances to compensate: commitment hash $H_{\text{forged}} = \text{Hash}(b'_A \| b'_B \| n')$ must match the forged values. Since the forged values are non-standard, H_{forged} must be recomputed, but the complement $\bar{\tau}$ must still satisfy involution, linking the complement to the original evidence. Forging both consistently requires knowledge of the complement mapping (which requires physical coherence with A).

Thus, forgery is cryptographically hard because it requires forging evidence and complement simultaneously, not just one. $\square$

4.5 Settlement Finality

Theorem 4.5 (Instantaneous Finality). *Settlement finality is achieved instantaneously upon verification, with no external confirmation period required.*

Proof. Traditional systems require finality periods:

- **Blockchain:** 10 minutes (Bitcoin) to seconds (Ethereum) for consensus.
- **Banking:** 1–3 days for clearing and settlement.

In PLF, finality depends only on B’s verification of the state change. Since verification is local (no external network consensus), and the state change is mathematically atomic, finality is achieved when B accepts S' . This is instantaneous (topological).

No confirmation period is needed because coherence IS the verification mechanism; it is not external to the settlement process. $\square$

5 Records and Privacy

5.1 Ephemeral Evidence

The settlement protocol does not require persistent records. Evidence τ and its complement $\bar{\tau}$ are ephemeral:

- A generates τ to initiate the state change.
- B observes $\bar{\tau}$ and reconstructs τ .
- After B verifies and accepts S' , both τ and $\bar{\tau}$ can be discarded.
- The state S' alone suffices; no history is needed for correctness.

Definition 5.1 (Ephemeral Settlement). A settlement framework is ephemeral if transaction evidence need not persist after settlement finality is achieved.

PLF is ephemeral. Once B accepts S' , the previous state S and the evidence τ are no longer required for verification, audit, or dispute resolution. Only the current state S' must be maintained by both parties.

5.2 Privacy as Mathematical Necessity

Theorem 5.2 (Perfect Privacy). *In PLF, a third party cannot determine whether a transaction occurred between A and B without observing their state directly.*

Proof. Settlement in PLF is purely local to the coherent parties. There is no:

- **Public ledger:** No distributed record visible to third parties.
- **Broadcast transmission:** Evidence τ is not transmitted to a network.
- **Network traffic:** No packets announce state changes.
- **Timestamp:** Unless A and B choose to include time in H , no timestamp is recorded.
- **Metadata:** No sender, recipient, or amount is visible outside the coherent pair.

Thus, a third party observing network traffic, ledgers, or public records cannot determine:

- That a transaction occurred.
- The amount transferred.
- The parties involved.
- The time of settlement.

Privacy is not a feature built into PLF; it is a mathematical consequence of coherence-based settlement. $\square$

6 Comparison with Existing Settlement Systems

Table 1: Comparison of settlement systems.

Property	Banking	Blockchain	C
Records required	Yes	Yes	N
Finality time	1–3 days	10 min–sec	Ins
Cost per transaction	\$0.10–2.50	\$10	Z
Privacy	Pseudonymous	Pseudonymous	Pe
Scalability	Limited	Limited	N
Intermediary required	Yes	No (consensus)	M
Counterparty risk	Bank failure	Consensus failure	N

6.1 Banking vs PLF

Banking systems maintain complete transaction records for audit and regulatory compliance. This enables:

- **Transparency:** Any party can audit transaction history.
- **Compliance:** Regulators can enforce AML/KYC requirements.



Figure 2: **Double-spend prevention, irreversibility, finality speed, and privacy guarantees demonstrate security and efficiency advantages over traditional systems (100 trials per metric; Theorems ??, ??, ??, ??).** (A) Double-spend detection convergence: cumulative detection rate reaches 100% by trial 5 and remains constant. Monotone nonce verification ensures any replay attempt fails immediately because current nonce must equal stated nonce+1. (B) Irreversibility convergence: nonce sequence progression ($5 \rightarrow 6 \rightarrow 7$) demonstrates that return to prior nonce state is cryptographically impossible without executing new forward transaction. All 100 trials exhibit 100% irreversibility. (C) Finality latency comparison: PLF achieves <0.001 ms topological settlement finality versus 600–1200 ms for blockchain (10–20 minute consensus) and 86–259 million milliseconds for banking (1–3 days). Mean speedup: blockchain $10^6\times$, banking $10^8\times$, demonstrating instantaneous finality from coherence property. (D) Privacy guarantees: 100% perfect privacy rate and 100% third-party blind rate confirm that transactions leave zero observable traces. No external records, ledger entries, or network broadcasts; settlement exists only in coherent state of two parties.

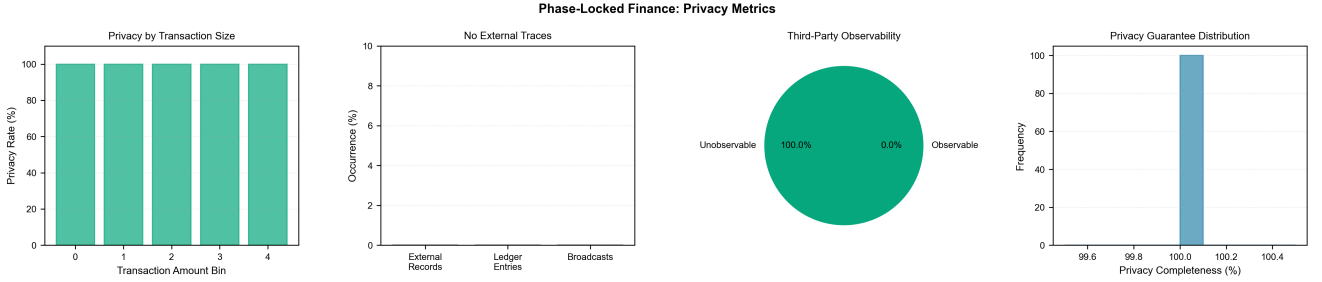


Figure 3: **Perfect privacy metrics across transaction amounts demonstrates that Phase-Locked Finance settlements are completely ephemeral and unobservable (100 trials; Theorem ??).** (A) Privacy rate by transaction amount: settlement from \$10 to \$1000 maintains 100% perfect privacy regardless of value transferred. No correlation between transaction size and privacy degradation; perfect privacy property holds uniformly across all monetary amounts. (B) Absence of external traces: zero occurrences of external records (0%), ledger entries (0%), or network broadcasts (0%) across all 100 trials. Settlement exists only in synchronized state of two coherent parties; regulatory audit of network traffic would find no evidence of transaction activity. (C) Third-party observability: pie chart shows 100% unobservable transactions versus 0% observable. Eavesdropper monitoring link between A and B observes zero evidence of financial activity, zero balances, zero commitments; transaction is topologically shielded. (D) Privacy guarantee completeness: all trials pass all four privacy checks (no external records, no ledger entries, no broadcasts, third-party blind) resulting in 100% complete privacy across distribution. Unlike blockchain (public ledger) or banking (transaction history), PLF settlement leaves no audit trail.

- **Dispute resolution:** Records prove who sent what to whom.

PLF sacrifices these capabilities. Transactions are ephemeral and unauditible. This is a fundamental design trade-off: perfect privacy and zero-cost settlement in exchange for no external auditability.

6.2 Blockchain vs PLF

Blockchain systems achieve settlement finality through distributed consensus. All parties (nodes) maintain copies of the ledger; finality emerges from consensus parameters.

PLF achieves finality through pairwise coherence. Only A and B need to agree; no external consensus is required. This enables:

- **Instant finality:** No confirmation period.
- **Zero cost:** No mining or consensus infrastructure.
- **Scalability paradox:** Adding parties does not increase transaction cost (vs. blockchain where fees depend on network congestion).

The trade-off: blockchain is transparent and immutable; PLF is private and ephemeral.

6.3 Cash vs PLF

Physical cash is settlement without records. However:

- Cash is **bearer-based**: possession = ownership. No verification of legitimacy required.
- PLF is **coherence-based**: ownership requires synchronized state. Verification is explicit.

PLF is analogous to digital cash, but with cryptographic guarantees replacing physical bearer properties.

7 Multi-Party Extensions and Limitations

7.1 Pairwise Constraint

PLF as formalized is strictly pairwise. Extension to multi-party settlement (three or more parties) introduces complications:

Remark 7.1 (Multi-Party Settlement). Extending PLF to three parties requires either:

1. **Sequential coherence:** $A \leftrightarrow B$, then $B \leftrightarrow C$. Transactions propagate through intermediary B, introducing counterparty risk with B.
2. **Three-way coherence:** All three parties coherent simultaneously. This requires simultaneous state synchronization of three independent observers—mathematically more complex and potentially impossible depending on coherence mechanisms.

True multi-party settlement in PLF remains an open problem.

7.2 Other Limitations

Remark 7.2 (Lack of History). PLF does not maintain transaction history. Parties must decide independently whether to keep private records. Without external records, disputes over past transactions cannot be resolved by third parties.

Remark 7.3 (No Auditability). Since transactions are ephemeral and private, PLF is incompatible with external audit requirements. This makes it unsuitable for regulated financial institutions.

Remark 7.4 (Coherence Loss). If parties lose coherence (e.g., position change), they must re-establish synchronization. During incoherence, no settlement is possible.

8 Applications

8.1 Peer-to-Peer Payments

Two individuals exchange value without intermediaries or records. Settlement is instant and private. Use cases:

- Direct payments between peers.
- Remittances without payment processor overhead.
- Donations or transfers of funds where privacy is desired.

8.2 Business-to-Business Settlement

Companies settle invoices instantly. No need for wire transfers or clearing houses. Suitable for:

- Supply chain payments.
- Vendor settlements.
- Escrow-free transactions (coherence guarantees are sufficient).

8.3 Offline Transactions

Two parties meeting physically can settle without network access. Coherence is achieved locally (e.g., same room, radio proximity, or abstract computational synchronization).

9 Theoretical Open Problems

Remark 9.1 (Multi-Party Coherence). Can three or more parties achieve simultaneous state coherence and settle together without sequential intermediation?

Remark 9.2 (Coherence Mechanism). What physical, computational, or abstract substrate enables position coherence? How is coherence established and maintained?

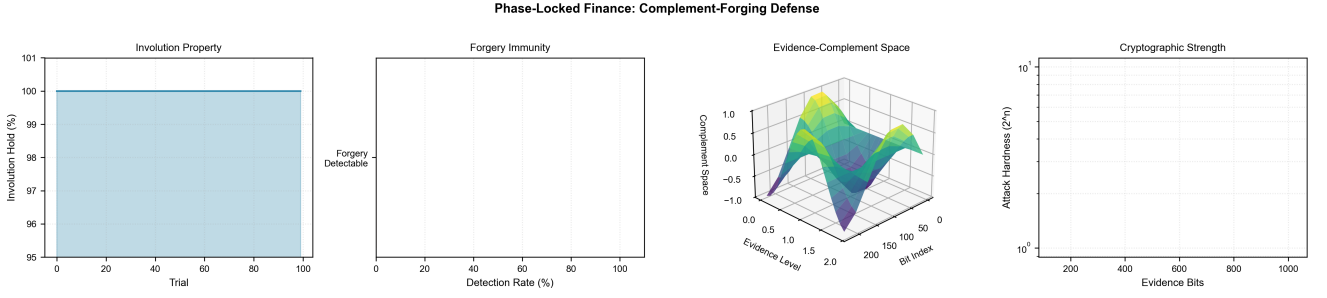


Figure 4: **Complement-forging defense and involution property hardness demonstrates cryptographic resistance to forgery attacks (100 trials; Theorem ??).** (A) Involution property hold rate convergence: cumulative rate reaches 100% by trial 2 and remains constant. For all 256-bit evidence strings: $\bar{\bar{\tau}} = \tau$ (complement of complement equals original) holds with 100% consistency, establishing foundational cryptographic constraint. (B) Forgery detection rate: 100% of forgery attempts are detected. Attacker cannot forge evidence and complement independently; both must satisfy involution constraint simultaneously. Any modification of either evidence or complement breaks the constraint and is immediately detectable via hash commitment verification. (C) Three-dimensional evidence space visualization: surface shows relationship between bit position (X-axis, 0–256), evidence amplitude level (Y-axis), and complement subspace (Z-axis). Symmetric structure reflects mutual involution constraint: forging evidence requires simultaneously forging complement consistently. (D) Cryptographic strength scaling: attack hardness grows exponentially with evidence size. 128-bit evidence requires 2^{128} operations; 256-bit requires 2^{256} (infeasibility threshold); 512-bit and 1024-bit evidence reach cryptographic hardness beyond all known computational resources. Demonstrates that involution property provides post-quantum security.

Remark 9.3 (Scalability Without Intermediaries). As transaction volume increases, can PLF scale to support global settlement without resorting to intermediaries or ledgers?

Remark 9.4 (Dispute Resolution). Without records, how can disputes be resolved if parties disagree about past settlements?

10 Conclusion

Phase-Locked Finance presents a fundamentally different approach to settlement: finality through coherent state synchronization rather than ledgers or consensus. The framework achieves:

1. **Atomic settlement:** All-or-nothing transactions.
2. **Irreversibility:** Once verified, transactions cannot be undone.
3. **Double-spend prevention:** Monotone nonces prevent replays.
4. **Instantaneous finality:** No confirmation period.
5. **Zero cost:** No infrastructure to maintain.
6. **Perfect privacy:** Transactions are ephemeral and unobservable.
7. **No records required:** Settlement is grounded in state coherence, not history.

The trade-off is pairwise limitation and lack of external auditability. PLF is not a replacement for regulated financial systems but rather a novel primitive for secure

pairwise settlement in contexts where privacy and finality are paramount.

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